

SENAATH SURaweera

Computer Science Undergraduate

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Profile

Motivated Computer Science undergraduate with a strong foundation in software development, problem-solving, and system design. Experienced in developing full-stack applications and academic projects using modern technologies. Passionate about building scalable solutions and continuously learning emerging technologies.

Education

University of Colombo School of Computing B.Sc. (General) in Computer Science	<i>Present</i>
Informatics Institute of Technology B.Sc. (Hons) in Computer Science	<i>Present</i>
Mahanama College, Colombo GCE A/L (Mathematics): 2A, 1B GCE O/L: 9A	<i>2009 – 2023</i>

Projects

RideMachan (Java, JavaScript, HTML/CSS)	GitHub
- Built a full-stack ride request management web application where users can submit requests and drivers can accept and manage them. - Implemented backend logic in Java with a clean HTML/CSS/JavaScript frontend interface.	
Recursive Descent Parser with Integrated Interpreter (C)	GitHub
- Hand-wrote a recursive descent parser and interpreter in C without any parser-generator tooling. - Parses and evaluates arithmetic and logical expressions by walking a grammar rule set.	
Get APOD (Java, JavaFX)	GitHub
- Developed a JavaFX desktop application that fetches and displays NASA's Astronomy Picture of the Day via the NASA public API. - Supports date-based browsing with image display and metadata (titles, explanations).	
Edmonds-Karp Algorithm (Java)	GitHub
- Implemented the Edmonds-Karp algorithm for maximum network flow using BFS-based augmenting path selection. - Achieves $O(VE^2)$ complexity with support for directed weighted graphs.	
Dinic's Algorithm (Java)	GitHub
- Implemented Dinic's algorithm constructing level graphs via BFS and finding blocking flows using DFS. - Achieves $O(V^2E)$ complexity, outperforming Ford-Fulkerson on dense graphs.	
Middleware Architectures (Java)	GitHub
- Explored common middleware patterns including message brokers, RPC, and event-driven communication. - Each module demonstrates a different middleware style with practical, runnable examples.	
Ticket Simulation (Java)	GitHub
- Built a multi-threaded ticket booking simulation modelling concurrent vendor and customer threads competing for a shared ticket pool. - Demonstrates synchronization, thread safety, and producer-consumer patterns.	

Advanced Ludo Simulation (Java, LWJGL, OpenGL)

GitHub

- Built a fully functional Ludo game using LWJGL and OpenGL, featuring four AI-driven players competing autonomously on a rendered graphical board.
- Implemented a game loop, OpenGL rendering, dynamic movement logic, mystery cells with random effects, and rule-based AI decision-making.

Plane Ticket Booking System (Java)

GitHub

- Designed and implemented a seat reservation system using Object-Oriented Programming principles.
- Handled booking logic, seat allocation, and user data management.

Real-Time Chat Application (Spring Boot, JavaFX, WebSockets)

GitHub

- Built a multi-user real-time chat system with a Spring Boot backend and JavaFX client, enabling simultaneous connections and instant message delivery via WebSockets.
- Implemented live message broadcasting, user session handling, and concurrent client support.

Bookstore API (Java, JAX-RS)

GitHub

- Developed a RESTful API for a bookstore system using JAX-RS, supporting CRUD operations for books, customer orders, and inventory management.
- Designed structured endpoints following REST principles with JSON-based communication.

2D Arcade Shooter (Java)

GitHub

- Built a 2D shooter game from scratch without a game engine.
- Implemented collision detection, scoring systems, and real-time gameplay logic.

Student Progression Predictor (Python)

GitHub

- Developed a system to predict student outcomes using data analysis techniques.
- Visualized results using histograms for better interpretation.

To-Do Application (C / Python)

GitHub

- Created a task management system using linked lists and file storage (CSV).
- Enabled efficient data handling and persistent storage.

Technical Skills

Programming: Java, C, C++, Python

Web Development: HTML, CSS, JavaScript, React

Backend & Databases: Spring Boot, Django, SQL

Tools: Git, GitHub, MySQL, Notion, ClickUp

Core Competencies

- Problem Solving & Critical Thinking
- Teamwork & Leadership
- Communication Skills
- Time Management & Adaptability

Extracurricular Activities

- President – Astronomy Society, Mahanama College
- President – Buddhist Society, Mahanama College
- Member – Rotaract Club of Colombo West
- Member – ICT Society
- Student Parliament Member & Junior Prefect

Interests

Music, Weightlifting, chess

References

Prof. A.M. Attygalla

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Shehan Kumar

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